

Math 125

Integration and Applications

Instructor: Gaku Liu (he/him) gakuliu@uw.edu

TA's: Haocheng Cai
Jonas Golm
Bryan Lu

Clarice Maciel
Connor Thomas McCawland
Jenny Zhan

Office hours TBA.

Lectures MWF

Quiz sections TTh

More info on Canvas syllabus

"Gaku"
"Professor Liu"

E-textbook: **Calculus: Early Transcendentals, 9th ed**

Purchase by clicking the "**Cengage WebAssign**" link on Canvas.

Required to access WebAssign, which contains homework.

My recommendation is to read each textbook section before the corresponding lecture, according to the schedule found on the Math 125 materials page.

Many homework problems are based on textbook examples, so it is a good place to look if you are stuck.

AI policy:

AI usage on homework is not forbidden, but strongly discouraged. Homework is only 5% of your grade, and relying on AI will leave you unprepared for quizzes and exams, which are the vast majority of your grade.

Learning is a physical process. You cannot learn math by watching, just as you can't learn to play an instrument by listening to music.

True or False: Integration is the opposite of Differentiation.

Partially true, partially false.

The opposite of differentiation is called antidifferentiation.

This is a **method** for computing integrals.

But integration is a separate concept, and there are many other ways to compute integrals.

True or False: Math is hard.

True.

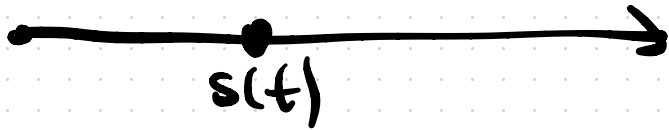
No one is born good at math.

Antiderivatives

(4.9)

Recall that the **derivative** of a function $f(x)$ is a function $f'(x)$ that outputs the instantaneous rate of change of f at a value x .

Example: Let $s(t)$ be the position of a particle^u at time t . moving in a line



$$s'(t) = v(t) \quad \text{velocity}$$

$$s''(t) = v'(t) = a(t) \quad \text{acceleration}$$

In your previous calculus class, you discussed how to take derivatives of some common functions.

$$f(x) = 2x^2 + 3x + 1$$

$$f'(x) = 4x + 3$$

$$f(x) = \sin x + \sqrt{x} + \ln x$$

$$f'(x) = \cos x + \frac{1}{2\sqrt{x}} + \frac{1}{x}$$

Let $F(x)$ be a function. Suppose we know the derivative of $F(x)$ is $f(x)$, where

$$f(x) = x^2$$

What is $F(x)$?

$$\begin{aligned} F(x) &= \frac{1}{3}x^3 & \text{or} & & F(x) &= \frac{1}{3}x^3 + 1 \\ & & \text{or} & & F(x) &= \frac{1}{3}x^3 - 1 \\ & & \text{or} & & F(x) &= \frac{1}{3}x^3 + 99.9 \\ & & \text{or} & & F(x) &= \frac{1}{3}x^3 + \frac{\pi}{\sqrt{2}} \end{aligned}$$

The functions $F(x) = \frac{1}{3}x^3$, $F(x) = \frac{1}{3}x^3 + 1$, $F(x) = \frac{1}{3}x^3 + 9.9$ are all **antiderivatives** of $f(x) = x^2$. In fact, any function of the form

$$F(x) = \frac{1}{3}x^3 + C,$$

where C is a real number, is an antiderivative of $f(x) = x^2$. The expression $F(x) = \frac{1}{3}x^3 + C$ is called the **general form** for the antiderivative of $f(x) = x^2$.

It is not a single function, but an **infinite family** of functions.

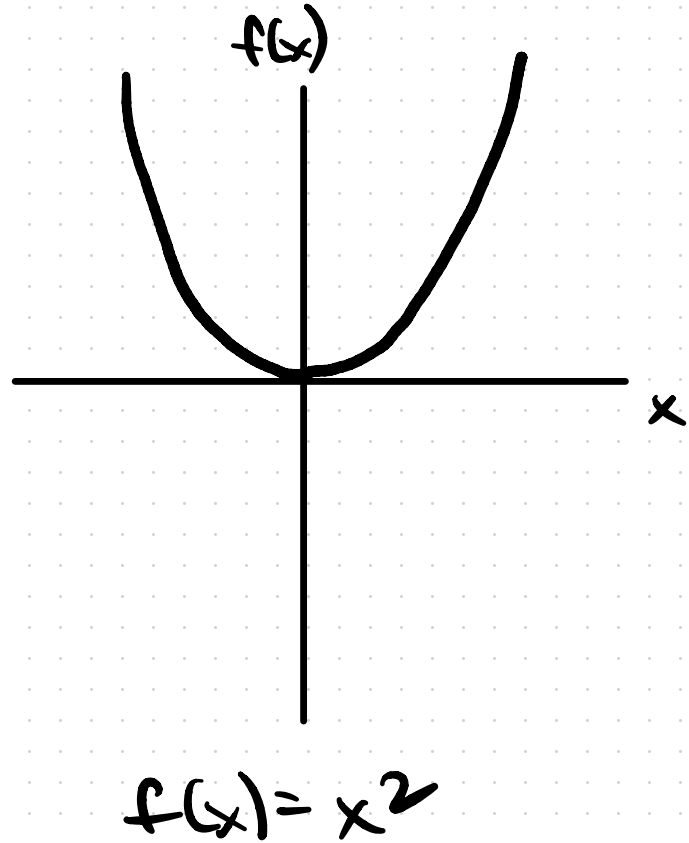
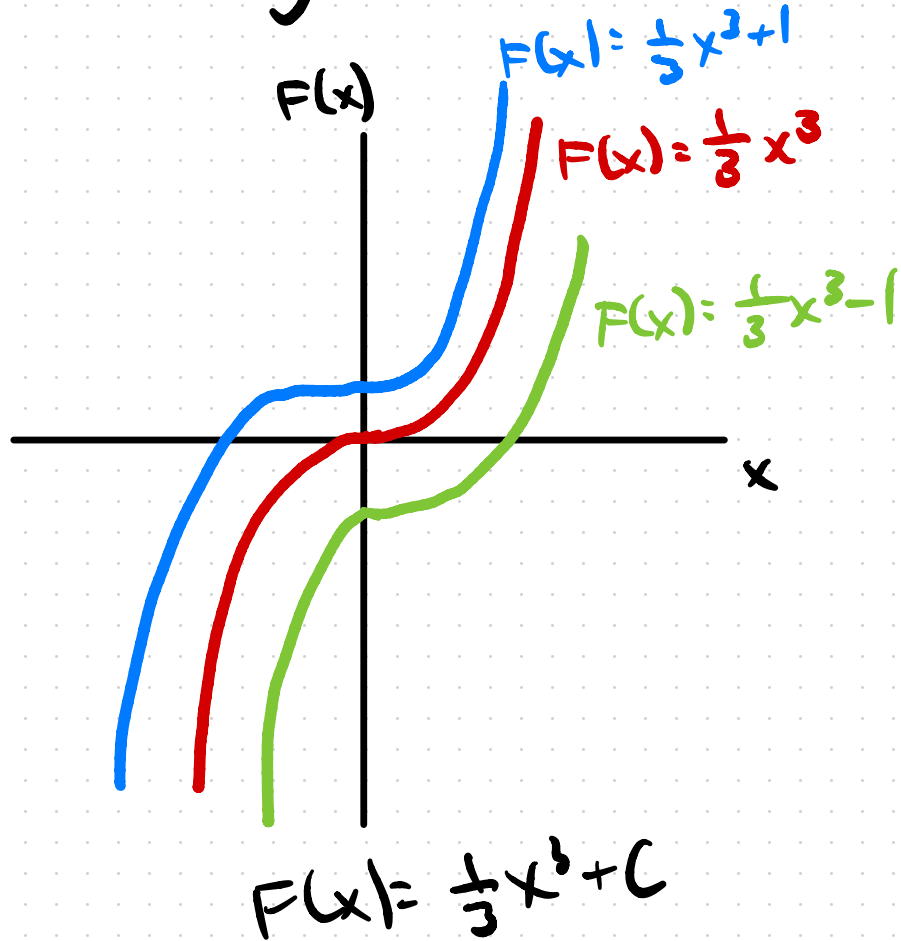
The following theorem tells us that the above "general form" actually covers all antiderivatives.

Theorem: Let $f(x)$ be a function defined on an interval, and let $F(x)$ be **any** antiderivative of f on that interval. Then **all** antiderivatives of f on that interval are of the form.

$$G(x) = F(x) + C$$

where C is a real constant.

Visualizing all the antiderivatives of $f(x)=x^2$.



Find the general form for an antiderivative of

$$f(x) = 5x^3 + \sin x + 4$$

$$F(x) = \frac{5}{4}x^4 - \cos x + 4x + C$$

Find the general form for an antiderivative of

$$f(x) = \frac{x^2 + x + 1}{\sqrt{x}}$$

$$= \frac{x^2}{\sqrt{x}} + \frac{x}{\sqrt{x}} + \frac{1}{\sqrt{x}}$$

$$= x^{3/2} + x^{1/2} + x^{-1/2}$$

$$= \frac{2}{5} x^{5/2} + \frac{2}{3} x^{3/2} + 2x^{1/2} + C$$

Suppose $f'(x) = 2x^2 + x$ and $f(0) = 1$. What is $f(x)$?

Start by finding the general form for $f(x)$.

$$f(x) = \frac{2}{3}x^3 + \frac{1}{2}x^2 + C$$

Plug in $x=0$, since we are given $f(0)=1$.

$$\begin{aligned} f(0) &= \frac{2}{3} \cdot 0 + \frac{1}{2} \cdot 0 + C \\ 1 &= C \end{aligned}$$

$$f(x) = \frac{2}{3}x^3 + \frac{1}{2}x^2 + 1$$

Suppose $g''(x) = x^2$, $g'(0) = 1$, $g(0) = 2$. What is $g(x)$?

Find the general form for $g'(x)$

$$g'(x) = \frac{1}{3}x^3 + C$$

Find the general form for $g(x)$

$$g(x) = \frac{1}{12}x^4 + Cx + D$$

D is a constant
different from C .

$$g'(0) = 1, \quad g(0) = 2$$

Plug in $x=0$ to expressions for g' and g

$$g'(0) = C$$

$$1 = C$$

$$g(0) = 0 + C \cdot 0 + D$$

$$2 = D$$

$$g(x) = \frac{1}{12}x^4 + x + 2$$